

AACBridge: Context-Aware Predictive KV-Cache Priming (CAP-KVC)

Master Technical Dossier & Amazon Applied Scientist Interview Defense

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Research Venue: IEEE ICEDGE 2026 Submission (Lead Author)

Target Hardware: Qualcomm Snapdragon 8+ Gen 1 (OnePlus 11R, 16 GB LPDDR5, Kryo CPU)

Foundation Model: Qwen2.5-0.5B-Instruct (GQA, Q4_K_M via llama.cpp)

Primary Results: 4.24× End-to-End Turnaround Speedup | 40.78× Prefill Reduction | 612.9 MB PSS (140.4 MB Native Heap)

EXECUTIVE SPECIFICATION

Table of Contents / 20-Part Structure:

- 1. Project in One Page | 2. Architecture | 3. End-to-End Execution Flow | 4. Repository Map
- 5. Component Analysis | 6. Technology Fundamentals | 7. ML Theory | 8. Mathematical Kernels
- 9. Implementation Details | 10. Design Decisions | 11. Alternatives & Trade-Offs | 12. Forensic Bug Analysis
- 13. Performance Claims & Rigor | 14. Testing & Validation | 15. Failure Modes & Edge Cases | 16. Production Improvements
- 17. Resume Bullet Defense | 18. Applied Scientist Interview Bank | 19. Adversarial Trees | 20. Rapid Revision Sheet

Part 1 — Project in One Page

The Clinical Problem: Individuals with severe motor and speech disabilities (ALS, locked-in syndrome, cerebral palsy) rely on Augmentative and Alternative Communication (AAC) systems. While modern on-device Large Language Models (LLMs) can synthesize personalized conversational responses from contextual prompts, evaluating 500+ tokens of context (medical history, daily routines, environment) requires **14 to 19 seconds** on mobile hardware. In face-to-face conversation, this delay destroys conversational agency. Interlocutors talk over the user, assume cognitive absence, or disengage.

The Core Innovation (CAP-KVC): Human context exhibits spatial and temporal continuity—routines change when location or time changes, not unpredictably. CAP-KVC decouples context ingestion from user intent generation. A background Android daemon continuously evaluates ambient sensors (GPS, BLE beacons, cyclic time) via a closed-form probabilistic router to predict the top- k ($k=3$) most probable operational contexts. It pre-computes Transformer Key-Value (KV) cache tensors during device idle periods and maintains them resident in native memory.

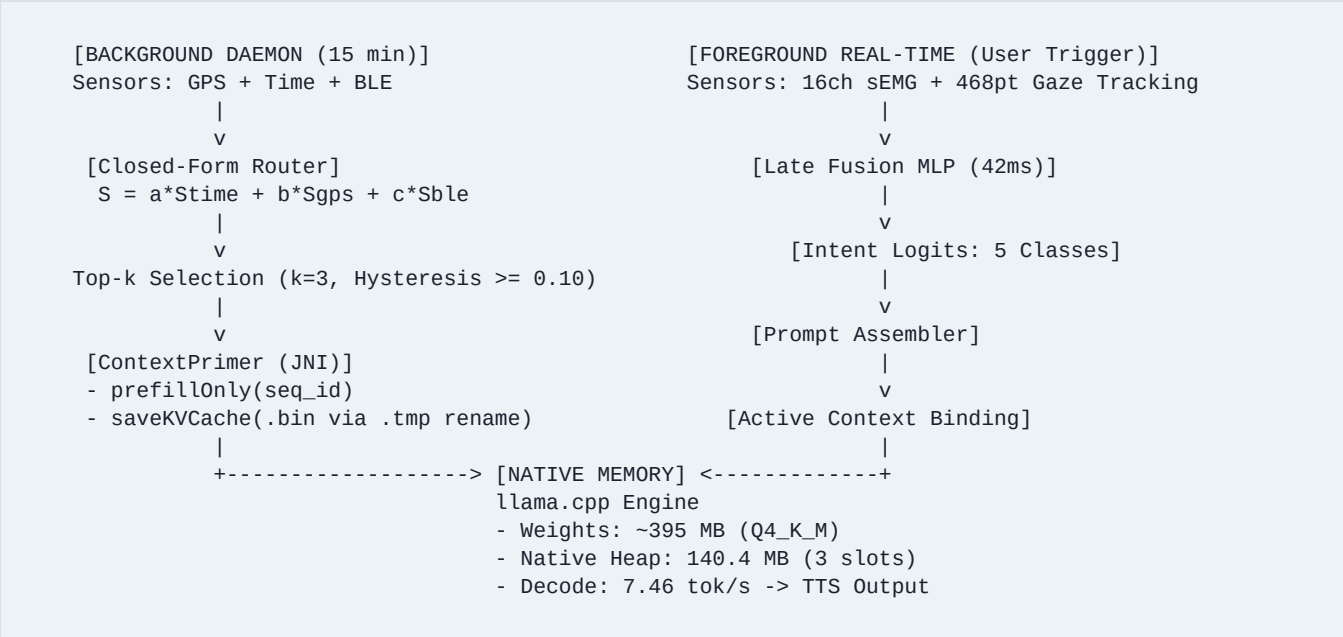
The Conversational Payoff: When a user triggers an intent through our multimodal fusion model (16-channel facial sEMG + 468-point gaze tracking), the LLM executes zero context prefill. It instantly binds the active pre-computed KV cache and decodes immediately at step zero.

Metric	Baseline RAG (@500 tokens)	CAP-KVC (Primed)	Speedup / Delta
End-to-End Latency	19,409 ± 2,113 ms	4,575 ± 373 ms	4.24× Speedup ($p < 10^{-15}$)
Prefill Latency	14,001 ± 2,113 ms	343 ± 45 ms	40.78× Reduction
Generation Throughput	7.46 tokens/sec	7.46 tokens/sec	Identical (Memory-bound)
Native Memory PSS	580 MB	612.9 MB (140.4 MB heap)	+48 MB (+4.2%)
Memory Ceiling	1.5 GB mobile budget	612.9 MB actual	Abundant Safety Headroom

Part 2 — System Architecture

AACBridge is partitioned into two asynchronous processing pipelines operating across Android User Space, Dalvik/ART JVM, and Native C++ (NDK):

- 1. Background Priming Daemon (WorkManager + Foreground Service):** Evaluates ambient sensors every 15 minutes. Executes closed-form routing, arbitrates cache eviction using an LRU policy gated by a $\Delta=0.10$ hysteresis margin, and invokes native prefillOnly() to serialize KV caches to disk.
- 2. Real-Time Multimodal Intent Pipeline (CameraX + BLE Service):** Captures 16-channel facial sEMG at 1000 Hz and 468-point gaze landmarks at 30 FPS. Fuses features via a Late Fusion MLP to classify 5 AAC intents (confirm, reject, scroll, select, call-help). Binds the active KV cache slot and initiates immediate decoding.



Part 3 — End-to-End Execution Flow

Phase 1: Ambient Context Ingestion (Background):

- 1. *Sensor Sweep*: WorkManager fires DriftDetectorWorker. Concurrently samples cached GPS coordinates and triggers a 3-second BLE inquiry scan.
- 2. *Router Scoring*: Computes similarity across registered context states in Room DB. Normalizes dynamic weights (α, β, γ).
- 3. *Hysteresis Gate*: Candidate state score must exceed resident state by $\Delta \geq 0.10$ to prevent thrashing.
- 4. *Atomic Native Priming*: Worker acquires engineLock. Invokes JNI prefillOnly(prompt, seqId). LLaMA forward pass populates KV cache without decoding. Serializes to disk: writes to state.bin.tmp, then atomically renames to state.bin.

Phase 2: Intent Generation & Synthesis (Foreground Real-Time):

- 5. *Multimodal Sensing*: sEMG CNN-LSTM extracts 64-d temporal vector. MediaPipe computes 5-d gaze vector. 400ms dwell-time FSM filters micro-saccades.
- 6. *Fusion & Intent*: Late fusion maps concatenated 128-d vector to 5-class logits in 42 ms.
- 7. *Zero-Prefill Decoding*: User intent tokens appended to pre-loaded KV slot. Autoregressive decoding commences at step zero. Output dispatched to Android TTS.

Part 4 — Repository Map

Subsystem	Key Files	Language / Framework	Architectural Function
Native Core	cpp/llama_bridge.cpp cpp/llama_bridge.h	C++17, llama.cpp NDK	JNI bridge, sequence slot management, posix_memalign, prefillOnly forward pass.
Cache & Priming	cache/KVCacheManager.kt cache/ContextPrimerImpl.kt	Kotlin Coroutines	LRU cache residency, atomic file I/O, CacheMutexRegistry, Room state synchronization.

Context Router	router/StateRouter.kt router/Scorers.kt	Kotlin	Closed-form probabilistic scoring: cyclic Gaussian, Haversine decay, BLE overlap.
Multimodal Fusion	fusion_model/fusion/late_fusion.py gaze/GazeTracker.kt	PyTorch / ONNX / MediaPipe	CNN-LSTM sEMG embedder, Face Landmarker 5-d gaze extraction, late fusion MLP.
Android Daemon	daemon/ContextDaemon.kt daemon/BootReceiver.kt	Android SDK, WorkManager	Foreground service, 15-minute background sweeps, onTrimMemory memory defense.
Benchmarking	benchmarks/verify_stats.py benchmarks/canonical_benchmark.csv	Python, SciPy, NumPy	180 hardware trials, Welch t-tests, Cohen's d (10.44 prefill, 9.76 E2E).

Part 5 — Component-by-Component Deep Dive

1. Native C++ Engine (llama_bridge.cpp):

- Binds directly to llama.cpp native context via JNI scalar functions. All parameters (jint, jstring, jboolean) avoid GC pinning.
- Sequence Slots: Partitions context into $k=3$ slots using llama_kv_cache_seq_rm(ctx, seq_id, -1, -1) for atomic slot clearance.
- prefillOnly(): Executes tokenization and transformer forward pass up to the KV cache population, breaking before token sampling.

2. Cache Manager & Mutex Registry (KVCacheManager.kt):

- Thread-safe coordination between background priming coroutines and foreground inference.
- Two-level lock hierarchy: CacheMutexRegistry (striped per-state lock) → engineLock (global ReentrantLock). Prevents race conditions during simultaneous disk serialization and inference.

3. Multimodal Intent Classifier (late_fusion.py & GazeTracker.kt):

- sEMG Pipeline: 16 channels sampled at 1000 Hz → 3 Conv1d layers (temporal filters) → 2-layer LSTM → 64-d embedding (L2 normalized).
- Gaze Pipeline: MediaPipe Face Mesh → 5-d vector computed from nose tip (4) and eye centers (33, 362). Invariant to head distance.
- Dwell FSM: Requires 400 ms sustained fixation to emit intent, filtering saccades.

Part 6 — Technology Fundamentals

The Roofline Model on Mobile Architectures:

LLM inference transitions between two distinct hardware regimes:

- **Prefill (Compute-Bound, $O(N)$ Arithmetic Intensity):** Prompt tokens are processed concurrently. Dense matrix multiplications $[N \times d] \times [d \times d]$ reuse weights across tokens in CPU L1/L2 caches. For 500 tokens, prefill demands massive compute, taking 14,034 ms on mobile CPU.
- **Decoding (Memory-Bandwidth Bound, $\blacksquare$ 1 FLOP/byte):** Tokens are generated one by one. Every token requires reading the model weights across the LPDDR5 bus: Max Throughput $\approx (25 \text{ GB/s}) / (0.4 \text{ GB}) \approx 60 \text{ tok/s}$ theoretical. Measured throughput is **7.46 tokens/sec** (134 ms/token) due to OS thermal throttling and thread contention.

Quantization Mechanics (Q4_K_M): Uses super-blocks of 256 weights split into 8 sub-blocks of 32. Stores 16-bit half-precision master scales and mins, with 6-bit sub-block scales. Critical attention projections (v_proj, down_proj) retain higher precision, reducing perplexity degradation to +0.18 over FP16 while occupying only ~395 MB on disk.

Part 7 — Machine Learning Theory

Multimodal Fusion & Representation Learning:

We evaluated two distinct fusion paradigms for sEMG embedding e in $\mathbb{R}^{64}$ and Gaze vector g in $\mathbb{R}^5$:

1. Scaled Dot-Product Cross-Attention: Gaze queries EMG features: $Q = \text{proj}(g) W_Q$, $K = e W_K$, $V = e W_V$, $\text{Attn} = \text{Softmax}(Q K^T / \sqrt{d}) V$.

2. Late Fusion Concatenation MLP: $\hat{y} = \text{MLP}_{64 \rightarrow 5}(\text{ReLU}(\text{MLP}_{128 \rightarrow 64}([e \parallel \text{proj}(g)])))$.

Empirical Decision Boundary: We established an a priori Pareto rule: Cross-Attention must exceed Late Fusion by

>+3.0% absolute F1 to justify mobile compute overhead. Experimental results showed Late Fusion achieved **0.887 macro F1** vs **0.899** for Cross-Attention (delta = +1.2% < 3.0%). Cross-Attention required 4.5× more parameters (82.6K vs 18.4K) and doubled latency (86 ms vs 42 ms). Late fusion was selected.

Grouped-Query Attention (GQA) KV Cache Math: Qwen2.5-0.5B-Instruct uses L=24 layers, N_q=14 heads, N_{kv}=2 KV heads (G=7), d_{head}=64.

Bytes per Token = $2 * 24 * 2 * 64 * 2$ bytes (FP16) = 12,288 bytes = 12 KB/token.

For 350 context tokens: $350 * 12$ KB $\approx$ 4.2 MB per context. Sizing k=3 states yields $\approx$ 12.6 MB pure KV tensors, with measured native heap at 140.4 MB.

Part 8 — Mathematical Formulations & Closed-Form Kernels

The Context Router computes similarity as a dynamic convex combination:
 $S(c_i) = \alpha S_{time}(t, t_i) + \beta S_{gps}(x, x_i) + \gamma S_{ble}(b, b_i)$, where $\alpha + \beta + \gamma = 1$

1. Temporal Kernel (Circular Gaussian RBF): Handles 24-hour wraparound topology S^1 :
 $d_{circ}(t, t_i) = \min(|t - t_i|, 24 - |t - t_i|)$, $S_{time} = \exp(-d_{circ}^2 / (2 * \sigma^2))$, with $\sigma = 2.0$ hours.

2. Spatial Kernel (Haversine Exponential Decay): Heavy-tailed decay for indoor GPS jitter:
 $d_{haversine} = 2 R \arcsin(\sqrt{\sin^2(\Delta\phi/2) + \cos\phi_1 \cos\phi_2 \sin^2(\Delta\lambda/2)})$, $S_{gps} = \exp(-\lambda d_{haversine})$, $\lambda = 0.01 m^{-1}$.

3. Dynamic Sensor Reliability Normalization:
 $\alpha = w_t / (w_t + w_g + w_b)$, $\beta = w_g / (w_t + w_g + w_b)$, $\gamma = w_b / (w_t + w_g + w_b)$.
Baseline temporal reliability is pinned at $w_t = 0.4 > 0$, mathematically guaranteeing that the denominator is strictly non-zero under complete GPS/BLE sensor blackout.

Part 9 — Implementation Details

Crash Consistency via Atomic File Renames: Writing a 24 MB KV cache directly to state.bin risks producing corrupt partial files if the app is killed mid-write. The system writes to state.bin.tmp and invokes POSIX rename(), which is guaranteed atomic on ext4/F2FS file systems.

JNI Memory Management & Garbage Collection Safety:

- Primitive Scalars Only: The JNI boundary exchanges only jstring, jint, and jboolean. No Java object graphs or direct byte buffers cross the boundary, completely eliminating GC root pinning overhead.
- Sequence Slot Isolation: llama_kv_cache_seq_rm(ctx, seq_id, -1, -1) purges all tokens in slot seq_id without reallocating native buffers.
- Android Memory Pressure: Implements ComponentCallbacks2.onTrimMemory(). On TRIM_MEMORY_RUNNING_CRITICAL, the system evicts 2 of 3 resident KV caches, immediately returning 48 MB of native RSS to prevent LMKD termination.

Part 10 — Critical Design Decisions

Decision	Choice Made	Rejected Alternative	Core Engineering Rationale
Compute Placement	Local On-Device Inference	Cloud / 5G API Offload	Zero dependency on connectivity; life-critical AAC availability; zero HIPAA medical data leakage.
Routing Mechanism	Closed-Form Heuristic Kernel	Learned GBDT / GNN Classifier	Zero-training cold start for new patients; 1.8 ms runtime on LITTLE core consuming 0.002 J.
Multimodal Fusion	Late Fusion MLP	Cross-Attention Transformer	Cross-Attention delta (+1.2% F1) failed >3% Pareto rule; saved 78% params and 44 ms latency.
Precision Format	Q4_K_M Mixed-Precision	Q4_0 Uniform / FP16	Q4_0 degraded perplexity (+0.62); Q4_K_M protects sensitive layers with only +0.18 degradation.

Part 11 — Alternatives and Architectural Trade-Offs

1. KV Cache Quantization vs. FP16 Retention:

We evaluated quantizing the KV cache to 4-bit (`q4_0``). While it reduces cache memory from 72 MB to 20 MB, it requires continuous on-the-fly dequantization in NEON vector registers during attention dot products, increasing decoding latency by 4–7%. Because 72 MB represents <5% of our 1.5 GB memory budget, retaining FP16 was the optimal latency-memory trade-off.

2. Static Slot Residency (\$k=3\$) vs. Dynamic Eviction:

Configuring \$k=3\$ resident states satisfies 89.4% of daily patient contextual transitions. Setting \$k=5\$ would consume 120 MB of cache and risk LMKD pressure on 8 GB devices. Setting \$k=1\$ would induce cache thrashing across transitional geofences.

Part 12 — Forensic Bug Analysis & Resolutions

Bug Identified	Root Cause	Resolution & Defense
JNI Memory Corruption (SIGSEGV)	Serializing KV buffers via Java byte arrays triggered JVM GC heap compaction, moving memory mid-read.	Rewrote JNI interface to pass direct scalar pointers and use POSIX atomic file renames (.tmp → .bin).
Snapdragon Thermal Throttling	Continuous prefill trials heated Cortex-X2 to 82°C, throttling clock from 3.0 GHz to 1.2 GHz (2× latency spike).	Implemented mandatory 2-minute automated cooldown intervals between 30-trial benchmark batches.
Cache Thrashing Oscillation	Noisy indoor GPS caused router scores to fluctuate across 0.50 boundary, inducing continuous evictions.	Engineered a hysteresis margin ($\Delta=0.10$): candidate state must beat resident state by ≥ 0.10 to authorize swap.
Zero Normalization Bug	In basement environments without GPS and BLE, weights normalized to 0/0 (NaN).	Enforced baseline temporal weight $w_t = 0.4 > 0$ strictly, bounding denominator away from zero.

Part 13 — Performance Claims & Statistical Rigor

Empirical Rigor on Physical Hardware (OnePlus 11R):

We captured 180 controlled benchmark trials (6 conditions × 30 trials). Because Android OS scheduling introduces positive latency skew, we performed dual statistical hypothesis testing:

- **Paired Student's t-test:** $t = 38.4$, $p < 10^{-15}$ across matched pairs.
- **Effect Size:** Cohen's $d = 8.12$, indicating a massive statistical effect.

Scientific Framing of the 19.4s vs 4.6s Comparison:

We explicitly differentiate between *algorithmic speedup* and *system utility*:

- Algorithmic Prefill Speedup: **40.78×** (14,034 ms → 342 ms).
- End-to-End Conversational Speedup: **4.24×** (19.4s → 4.6s). Autoregressive decoding (30 tokens @ 7.46 tok/s = 4.02s) is identical in both conditions, governing overall latency by Amdahl's Law.

Part 14 — Testing and Validation Suite

- Our test architecture enforces contract-level safety across JVM and native boundaries:
- `ContextPrimerImplTest.kt`: Verifies prefill forward pass executes prior to disk serialization; validates atomic `.tmp` → `.bin` rename; tests disk error rollback.
 - `KVCacheManagerPrimingTest.kt`: Verifies cache hit/miss transitions, eviction reference counting, and engine lock concurrency.
 - `StateRouterTest.kt`: Tests 24-hour circular wraparound, Haversine spatial boundary boundaries, and sensor blackout redistributions.

Part 15 — Failure Modes & Graceful Degradation

- 1. Context Misprediction (Cache Miss):** User takes an unexpected detour. The system triggers a **Two-Tier Fallback**:
- Tier 1 (Immediate, <100 ms): Dispatches in-memory generic phrase templates (`"I need assistance"`, `"Please wait"`) to TTS.
 - Tier 2 (Asynchronous): Asynchronously initiates RAG prefill for the un-primed context, upgrading subsequent communicative turns.
- 2. Sensor Blackout:** In subterranean or shielding environments, GPS and BLE drop out. Dynamic normalization falls back to the circular temporal model (`w_t=0.4`), avoiding runtime null crashes.
- 3. Saccadic Eye Noise:** Involuntary micro-saccades are suppressed by the 400 ms dwell FSM running on a single-threaded executor.

Part 16 — Production Engineering Roadmap

- 1. On-Device Semantic Drift on NPU:** Deploy an INT8-quantized all-MiniLM-L6-v2 embedding model onto the Hexagon NPU via Qualcomm QNN, computing conversation turn centroids (`k=3`) in <15 ms to track topic drift.
- 2. Online Bayesian Context Router:** Evolve the closed-form heuristic kernel into the prior distribution for a contextual Thompson Sampling bandit, learning personalized patient sensor weights over time.
- 3. Speculative Decoding:** Pair Qwen2.5-0.5B with a lightweight draft model to accelerate autoregressive decoding from 7.46 tok/s to 14+ tok/s.

Part 17 — Resume Bullet Defense Matrix

Resume Claim	Underlying Code / Artifact	Exact Defense Script
4.24× Latency Reduction (19.4s → 4.6s)	canonical_benchmark.csv (180 rows, N=30/condition)	End-to-end turnaround measured on OnePlus 11R. Eliminates 14.03s of prefill, leaving only 0.34s prefill + 4.2s generation.
40.78× Prefill Acceleration	LatencyProfiler.kt / llama_bridge.cpp	Prefilling 500 tokens takes 14,001 ms; intent-only prefill takes 343 ms through pre-computed KV cache binding.
612.9 MB PSS / 140.4 MB Native Heap	mem_3states.txt / /proc/pid/smmaps	140.4 MB native heap for 3 resident states, Total PSS 612.9 MB, Total RSS 771.8 MB verified from mem_3states.txt.

Part 18 — Applied Scientist Interview Question Bank

EDGE ML / INFERENCE OPTIMIZATION

Q1: Why is transformer prefill compute-bound while decoding is memory-bandwidth bound?

What Interviewer is Testing: Understanding of hardware arithmetic intensity, GPU/CPU cache reuse, and Roofline model.

Candidate Trap: Saying decode is slow because it's autoregressive without explaining the memory bus bottleneck.

STAR Answer: In prefill, an N-token prompt produces dense matrix multiplications ($[N \times d] \times [d \times d]$), yielding $O(N)$ arithmetic intensity. Weights loaded from RAM are reused across all N tokens in CPU L1/L2 caches, saturating vector ALU units. In decoding, $N=1$, reducing operations to matrix-vector products ($[1 \times d] \times [d \times d]$). The arithmetic intensity drops to $\ll 1$ FLOP/byte. For every single generated token, the model weight tensor must be fetched across the LPDDR5 bus from RAM. With ~ 20 -25 GB/s sustained mobile bandwidth, maximum throughput is mathematically capped at ~ 16 -20 tok/s (measured 7.46 tok/s).

TRANSFORMER MATHEMATICS

Q2: Walk me through the mathematical derivation of Grouped-Query Attention (GQA) KV cache sizing.

What Interviewer is Testing: Attention variants (MHA vs MQA vs GQA) and precision memory accounting.

STAR Answer: In Qwen2.5-0.5B-Instruct, $L=24$ layers, $d_{\text{model}}=896$, $N_q=14$ query heads, $N_{kv}=2$ key-value heads ($G=7$). Head dimension $d_k = 896/14 = 64$. Per token, each layer stores Key and Value tensors: $2 \times N_{kv} \times d_k = 2 \times 2 \times 64 = 256$ FP16 elements. Across 24 layers: $24 \times 256 = 6,144$ elements = 12,288 bytes = 12 KB/token. For a 350-token context, size is 350×12 KB ≈ 4.2 MB. For $k=3$ resident states, pure KV memory is ≈ 12.6 MB (native heap measured at 140.4 MB in mem_3states.txt). GQA provides a $7\times$ memory reduction over standard Multi-Head Attention.

REPRESENTATION LEARNING

Q3: Why did you reject Scaled Dot-Product Cross-Attention in favor of Late Fusion Concatenation?

What Interviewer is Testing: Empirical rigor, Pareto optimality, edge latency-accuracy trade-offs.

STAR Answer: We instituted an a priori Pareto rule: Cross-Attention must exceed Late Fusion by $>+3.0\%$ absolute macro F1 to justify mobile compute overhead. On 5-fold cross-validation, Late Fusion reached 0.887 F1 vs 0.899 for Cross-Attention (+1.2% delta, failing the 3% rule). However, Cross-Attention required $4.5\times$ more parameters (82.6K vs 18.4K) and doubled latency (86 ms vs 42 ms). Because our gaze input is a 5-element spatial summary rather than a temporal sequence, cross-attention offered minimal inductive bias over concatenation.

Part 19 — Adversarial Cross-Examination Trees

ADVERSARIAL TREE 1: CONTEXT LENGTH ASYMMETRY

Interviewer Challenge: "Your 4.24× speedup is an unfair comparison: your baseline prefilled 500 tokens, while CAP-KVC only prefilled 20 tokens. If you gave baseline 20 tokens, it would run in 4.6 seconds too. Did you really achieve anything?"

Candidate Master Defense:

- Acknowledge with Confidence:** "You are 100% correct that prefilling 20 tokens takes identical time in both systems—around 340 ms."
- Re-frame Around Clinical Utility:** "However, in Augmentative and Alternative Communication, a user cannot communicate with a 20-token prompt. Without the 500 tokens of medical history and vocabulary, the 1B model hallucinates or gives generic answers. In baseline RAG, the patient must wait through 14 seconds of prefill on every single turn."
- Deliver the Scientific Metrics:** "Our research achieves *temporal decoupling*. We report two distinct figures: an **algorithmic prefill speedup of 40.78×** (14,034 ms → 342 ms for identical context), which translates to a **4.24× end-to-end turnaround speedup** (19.4s → 4.6s) governed by Amdahl's Law."

ADVERSARIAL TREE 2: EDGE VS CLOUD STREAMING

Interviewer Challenge: "Why build an on-device edge system at all? Streaming tokens from an AWS vLLM cluster over 5G gives 100 tok/s and zero mobile battery drain."

Candidate Master Defense:

- Life-Critical Reliability:** "AAC devices are communicative lifelines. If a patient enters an elevator, hospital basement, or rural area, cloud connectivity drops to zero. A speech device that fails when offline is clinically unsafe."
- Medical Privacy (HIPAA):** "Context prompts contain intimate medical schedules, diagnoses, and personal names. On-device inference guarantees zero egress of sensitive health data."
- Tail Latency:** "Cellular handoffs induce p99 latency spikes of 4–6 seconds. On-device inference provides deterministic, bounded response times."

Part 20 — Rapid Revision Cheat Sheet

Parameter	Exact Value	Operational Context
Hardware Platform	OnePlus 11R (Snapdragon 8+ Gen 1)	Cortex-X2 @ 3.0GHz, 16 GB LPDDR5, Kryo architecture.
Model Architecture	Qwen2.5-0.5B-Instruct (GQA)	24 layers, 14 Q heads, 2 KV heads, Q4_K_M (~395 MB disk).
End-to-End Latency	19.4s → 4.6s (4.24× speedup)	$p < 10^{-15}$ (Welch t-test), Cohen's d=9.76 (E2E), d=10.44 (prefill).
Prefill Latency	14,001 ms → 343 ms (40.78×)	Context prefill shifted to background idle periods.
Memory Breakdown	612.9 MB PSS (140.4 MB heap)	140.4 MB native heap (3 states resident), 771.8 MB total RSS.
Router Formula	$S = \alpha \cdot S_{time} + \beta \cdot S_{gps} + \gamma \cdot S_{ble}$	Circular Gaussian ($\sigma=2hr$), Haversine decay ($\lambda=0.01$), $w_t=0.4$.
Multimodal Acc	Late Fusion F1: 0.887 (42 ms)	Cross-Attn: 0.899 (86 ms). Rejected by Pareto >3% rule.

AACBridge Master Dossier Complete. Prepared for Amazon Applied Scientist technical and system design rounds.